TERN SYSTEMS

Memorandum

Configuration Editors Polaris Plan - Summer Project 2026

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Audience: Summer Interns

1 Background

When you have complex systems that need to be deployed to different customers you need system configuration, and quite a lot of it. In Polaris this is mainly done using xml or json files. For some system configuration, reading and modifying the files themselves can be fine, but for other files it can be quite difficult. This is especially true for files modifying how user interfaces are supposed to look, or configuration files which are using or modifying aeronautical data.

In previous years, we have started some work to create these kinds of configuration editors, mainly with a map layer editor tool and an aeronautical data editor. See presentations from previous summer projects: Show and Tell - Map Layer Editor and Aeronautical Data Editor Show & Tell.

The code for these projects is hosted in the following git repository: https://gitlab.com/TernDev/ATM/polaris-asd-editor

The aeronautical editor has been deployed to a customer in Hungary and will be deployed to a customer in Iceland soon as well as other future deployments. The map layer editor is still in the prototype stage but we're hoping to have time to incorporate it into deployments soon.

2 Goal

The aim for the summer is to create more configuration editor applications, each focused around a specific use case. The focus will be on UI configuration like flight list editing and track label editing. Creating an interface which can make it easy to create, visualize and update ASD[1] configurations. We might also look into creating editors for safety net configuration, which are tied to airspaces which would be good to visualize. Other options include aircraft performance editing and more depending on how progress is.

Flight list example:
https://documentation.tern.is/polaris-asd-user-manual/latest-upload/0000-polaris-ice-user-manual-intro.html#_sector_list_overview

Track label example:
https://documentation.tern.is/polaris-asd-user-manual/latest-upload/0000-polaris-ice-user-manual-intro.html#_track_representation_of_flight_states

3 Plan

3.1 Configuration file import/export

Create classes to pick out xml instances of a certain class for modification, and export them again while preserving the rest of the file contents. Considerations include what to do with configuration property attributes and how to use them in the editor. Here we would want to create a library of classes useful in all the new editors.

3.2 Field generation from XSD schemas

Automatically generating fields from XSD schemas can be a good candidate for some parts of the user interface to avoid having to update the editors each time slight modifications are made to the editors.

Exploration of generating fields from XSD schema was done in a recent hackathon by Hayfa Ayadi and can be found here:
https://gitlab.com/TernDev/sandbox/egillm/qt-wasm-exploration/-/tree/master/apps/schema-editor-generator?ref_type=heads

Furthermore, the entire aeronautical editor application is built up in this manner. Using the protobuf schema defining the aeronautical data to generate the user interface using predefined input fields.

[1] ASD is the Air Situation Display, the primary user interface for air traffic controls which shows flights on a map with supporting tools to control air traffic.

3.3 Create specific user interfaces for each use case

Create editors about each of the use cases previously mentioned, one for track labels, another for flight lists and hopefully more if time allows.

Exploration for this kind of track label editor was also done in a recent hackathon by Hayfa Ayadi and can be found here: https://gitlab.com/TernDev/sandbox/egillm/qt-wasm-exploration/-/tree/master/apps/track-label-editor?ref_type=heads

3.4 Compile to Web Assembly

Along with your work on the new editors, Egill Magnússon will also be looking into compiling the existing editors into Web Assembly, to make the applications accessible in any browser. If this proves successful, we will also apply this to your new editor applications in a similar manner.

See exploration hosted at: http://qt-webassembly-demo.tern.is/

4 Deliverables

You will be contributing code to the polaris-asd-editor repository and elsewhere as needed: https://gitlab.com/TernDev/ATM/polaris-asd-editor/

You will be expected to produce a final report summarizing your work along with a presentation for the company on the outcome of your work.